Research Framework



Pick markets → Define standards(Validate Regions)→ Users and Competitors → How to Land the Product

01

Regional Market Overview

Identify the regions and markets to research

02

Analytical Standards

Define the dual blue-ocean standards

03

Regional Market Analysis

Validate Korea / US / Germany one by one and find their Commonality

04

Trend Diffusion
(Suprising Finding)

Cross-market reference for launch sequencing

05

User Profile and Competitors

Needs, borrowings and pitfalls

06

UI Style Research

Cat theme × East-Asian anime style

07

Side Gameplay Research

Restaurant sim × match-3 gacha

08

Using which Element?

Sushi = the final sorting element

Conclusion

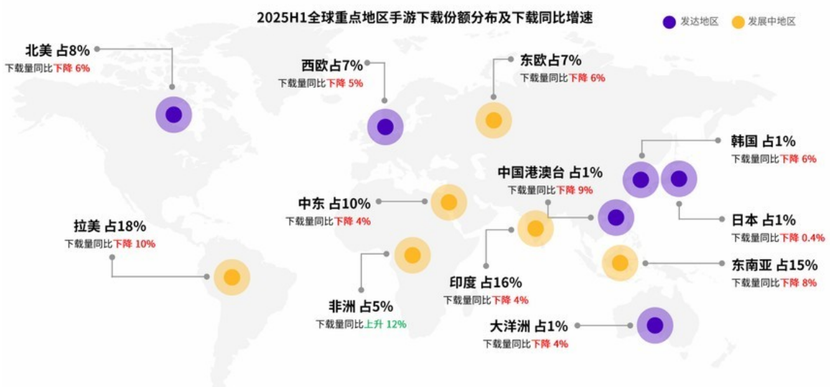
A development direction with **Sort gameplay at its core**
Cat Theme × Sushi Elements × Restaurant Simulation and Gacha Collection

Global Mobile Game Market

Based on 2025 H1 global market data, we lock in high-value regions across downloads, payments, and monetization

Download Share (by Region)

- User-acquisition growth is **weak** in most regions — a stock-market competition
- Africa grows against the trend on infrastructure and mobile-device adoption



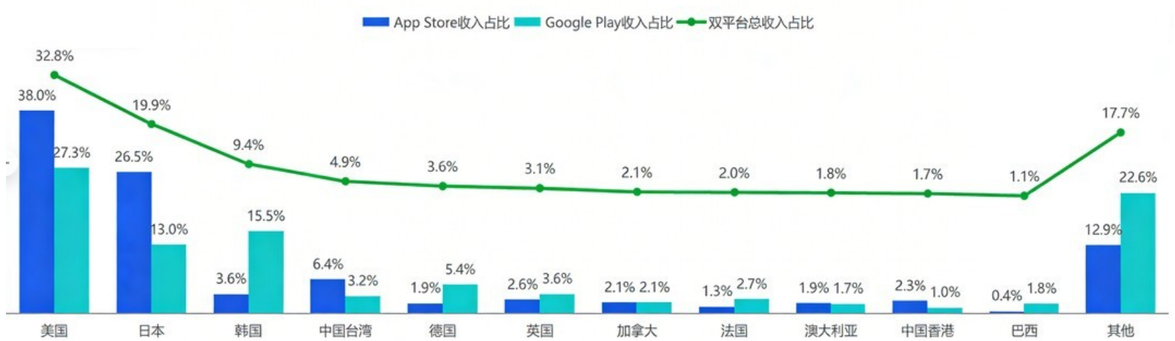


ARPD (Revenue per Download)

- Mature markets show far higher ARPD than emerging ones
- High-ARPD markets = the core IAP battlegrounds

Paying-User Ratio (by Region)

- Paying users concentrate in **mature markets**
- North America, East Asia and Western Europe are the payment core



Monetization (by Genre, Global)

- Pure-ad monetization is shrinking; **IAA ✦ IAP hybrid monetization** goes mainstream
- Casual genres are accelerating toward hybrid-casual



By market value, we selected **South Korea (Asia)**, the **United States (North America)**, and **Germany (Europe)** for deep research

Standard A: Game Chart Tracking

Track 90-day charts in each market to identify blue-ocean genres with weak competition, new demand, and sustainability

① Competition Intensity (HHI)

Metric: Sum of squared publisher chart shares

HHI **0.05** = fragmented, small teams have a chance; 0.15 = red ocean

② Demand Growth

Metric: Free-rank mean × grossing-rank mean

High free rank + low grossing rank = **traffic untapped by big publishers**

③ Genre Trend

Metric: 90-day rank slope across 3 segments

Negative slope = rank rising — **the genre is on an uptrend**

④ Stability

Metric: Days on chart + volatility

Longer presence and lower volatility = **real demand, long-run operable**

⑤ Paid-Chart Opportunity

Metric: Paid-vs-free rank gap

Positive gap = proven gameplay — **going free is the entry point**

⑥ Hit Freshness

Metric: Top-10 days on chart and first entry date

New entrants after day 45 = **the market is still evolving**

Composite Scoring Weights (normalized 0–100, weighted sum)

Traffic	Low Competition	Relative Stability	Low Entry Barrier	Trend
25%	25%	20%	20%	10%

Weighting logic: the goal is IAA + IAP monetization

Traffic and Competition Intensity are the core of blue-ocean judgment

Stability and barriers decide long-run operation

Trend only aids timing.

Sources: Newzoo, Global Games Market Report 2025; iyiou, 2026 China Mobile Game Overseas Insights; WeChat Mini Game Team, WeChat Open Class PRO 2026

Standard B: Industry Report Method

Per the team's definition and Naavik's "How Niche Subgenres are Reshaping the Mobile Puzzle Market" — a genre is a true blue ocean only when all three gates pass

Supply Side

CR3 30%

The first gauge of how "red" a red ocean is

Metric set

- Top-3 product revenue share (CR3)
- New launches per month / quarter
- UA competition intensity (CPI)

CR3 30% → blue-ocean candidate

Demand Side

YoY 50%

A blue ocean means new demand, not stealing existing users

Metric set

- Genre DAU YoY growth rate

DAU growth 50% → strong blue-ocean signal

Unit Economics / IAA-Friendly

CPI benchmark met

The business loop must close — otherwise it is false demand

Metric set

- CPI $0.61 (Android benchmark)
- CPI $4.09 (iOS benchmark)
- At least 3 IAA-friendly scenarios

UA cost below benchmark → loop closes

With **CR3 30%**, **YoY 50%**, and **IAA compliance**, we screen "true blue-ocean" subgenres in each regional market (see the regional analyses)

Reference: Naavik, "How Niche Subgenres are Reshaping the Mobile Puzzle Market"



Asia·South Korea (Macro Economy and Regulation)

그림 2-1-2-22 2024년 연구개발비

(단위 : %, n=816)

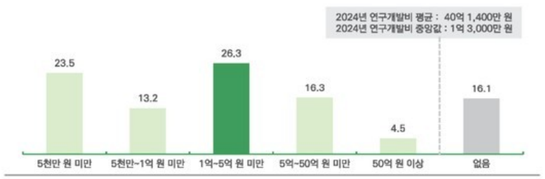


표 2-1-2-31 업체 특성별 2024년 연구개발비

(단위 : %)

구 분		사례수	5천만 원 미만	5천만~1억 원 미만	1억~5억 원 미만	5억~50억 원 미만	50억 원 이상	없음	평균 (백만 원)	중앙값 (백만 원)
전체		(816)	23.5	13.2	26.3	16.3	4.5	16.1	4,014	130
주력 플랫폼	PC 게임	(192)	20.8	14.6	27.1	17.7	4.2	15.6	1,350	122
	모바일 게임	(561)	24.4	13.5	25.3	15.9	5.2	15.7	5,311	129
	콘솔 게임	(21)	9.5	4.8	38.1	9.5	0.0	38.1	281	152
	아케이드 게임	(42)	31.0	7.1	31.0	19.0	0.0	11.9	410	134
사업 유형	게임 제작	(183)	23.0	9.8	21.9	12.6	1.1	31.7	377	100
	게임 배급	(44)	4.5	0.0	6.8	2.3	0.0	86.4	242	225
	게임 제작/배급	(589)	25.1	15.3	29.2	18.5	5.9	5.9	4,876	141
종사자 규모	5인 미만	(218)	40.8	12.8	13.3	2.3	0.0	30.7	123	30
	5인~9인	(192)	30.7	18.8	24.0	6.8	0.0	19.8	155	71
	10인~29인	(219)	18.3	16.9	43.4	13.7	0.0	7.8	298	134
	30인~99인	(96)	4.2	6.3	36.5	42.7	5.2	5.2	1,211	500
	100인 이상	(91)	0.0	1.1	11.0	48.4	35.2	4.4	29,160	2,597

주1) 사례 수가 과소한 집단(n<30)은 분석 대상에서 제외함.
주2) 그래프 및 도표 내의 숫자는 소수점 이하 둘째자리에서 반올림되었으므로 세부항목의 합이 합계와 불일치할 수 있음.

Tight regulation but strong support for small teams & highly mature internet infrastructure

New Game Regulations (densely enacted 2024–2025)

Policy	Effective	Key Points
Overseas Company Domestic Agent System	2025.10.23	Foreign publishers above scale without a Korean entity must appoint a local agent; fines up to KRW 20M
Mandatory Probability Disclosure	2024.03.22	Gacha / loot-box odds shift from self-regulation to legal duty, with punitive damages
"Dark Pattern" Ban	2025.03.20	Targets 6 manipulative designs: auto-renewal, hidden prices, pre-selected options, fake rankings, blocked cancellation
Rating Authority Delegation	2025.03.20	Non-teen ratings move to self-regulatory bodies — faster launch and updates for small games

Internet Infrastructure

50.6M internet users (**97.9%** penetration); 49.3M social-media identities

Industry Support: KRW 1 Trillion 'Content Industry Fund'

Backs console and indie development; micro-teams of ≤5 people get rating-fee reductions

The market still favours mid-core and hardcore games. In broad category terms, card games are a major blue ocean in the Korean market, while RPG, casual, and simulation games still hold considerable traffic and competitive opportunities.

Sources: IMF, World Economic Outlook; DataReportal, Digital 2026: South Korea; Chambers, Korea's Game Law 2025 Guide; MCST

North America·United States (Game Market and IAA Potential)

A steadily growing giant & Soaring UA costs force "stock operation + hybrid monetization"

$59–61B

2026E total game market output (+3.2–4.1%)

218M

players (65%+ of the population)

$325 / yr

spend per player; 12.5 hrs weekly

45%✦

mobile revenue share (incl. mini-games)

Spending Tiers: Small-and-Frequent ✦ High-Ticket Hybrid

- Casual mobile: **$25–40/yr** (steps, stamina — small, frequent buys)
- Hybrid-casual / mid-core: **$120–150/yr**, season passes and progression lift the cap
- 18–34 the largest cohort; 55+ silver gamers growing; 72% play cross-device

Competition: Small Teams Are the Supply Mainstay

- Indie developers **21%**; micro-teams of ≤5 are the largest group at **35%**
- **52%** of studios pivot to smaller, controlled projects to de-risk
- Top-5 publishers hold ~**39%** of engagement; non-US publishers take 45~55% of revenue (HungryStudio, Voodoo leading)

IAA Potential: The World's No.1 Mobile Ad Market

$174.49B

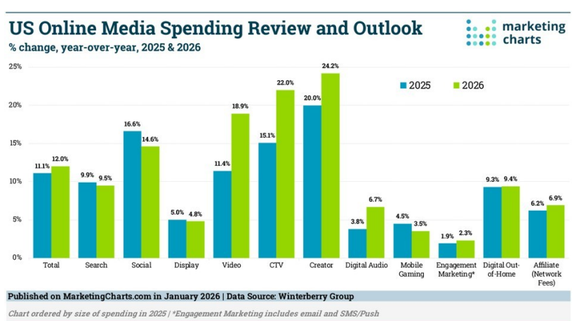
2025 digital ad market (CAGR 13–15%)

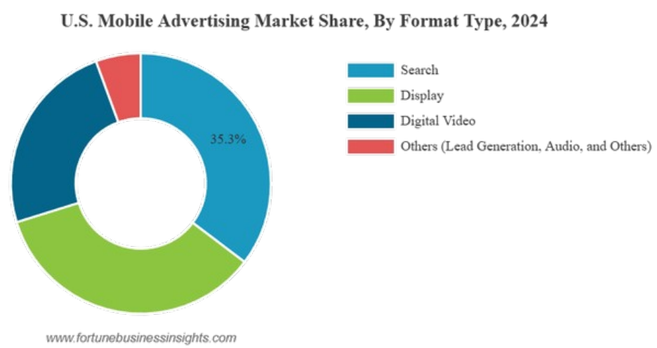
North America ~33.2% of global mobile ads

$11.03B

2025 in-game advertising (IAA) value

2030E: $13.7–20.7B





Trend: UA costs rose 30%+ in 2025–26, squeezing pure-IAA hyper-casual — the **hyper-casual → hybrid-casual transition** is the main theme

TikTok / AppLovin are the UA battlegrounds

Europe·Germany (Subgenre Validation and Chart Scores)

Finding Simulation / Puzzle as blue oceans

3D Spatial Sorting and Tidying

CR3 **12%**

DAU YoY **+58%**

Monetization **Strong IAP**

Spatial color-sort variants (fridge / makeup case / suitcase tidying); no long-term dominator

Visual Shelf-Sorting

CR3 **14.8%**

DAU YoY **+65%**

Monetization **Great hybrid**

3D goods matching + organizing; fragmented competition; balanced IAA and IAP

Screw-Metal Physics Sim

CR3 **21.5%**

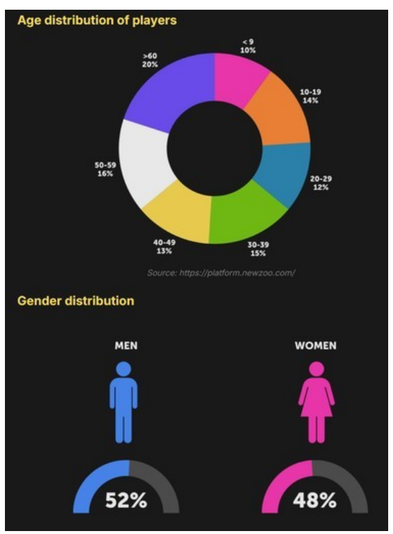
DAU YoY **+112%**

Monetization **High eCPM, low CPI**

Industrial-style Screw Away titles; a natural fit for German player taste

Composite Chart Scores (Google Play Germany, 90-Day Tracking)

Rank	Genre	Traffic	Low Comp.	Stability	Low Barrier	Trend	Total	Verdict
1	Simulation	50.3	89.4	58.7	60.3	50.8	63.8	Blue Ocean
2	Puzzle	74.9	30.9	54.2	100	40.4	61.3	Blue Ocean
3	Sports	28.1	100	56.7	40.0	74.6	58.8	Potential
4	Action	63.1	31.9	58.1	66.0	100	58.6	Potential
5	Strategy	78.5	30.1	69.0	49.2	65.2	57.3	Potential
6	Card	73.7	64.7	65.4	33.5	28.8	57.3	Potential
7	Casual	58.0	28.1	56.5	81.8	36.3	52.8	Potential
10	Role-Playing	74.4	65.8	34.6	24.5	0	46.9	Red Ocean



With an aging, high-willingness-to-pay player base, Simulation + Puzzle is the promising genre direction, shelf-sorting and screw physics are clear subgenre openings

Commonality —— Water Sort

As the only category of gameplay that meets the blue ocean standard in all regional markets, Water Sort is our top choice.

Competition Intensity

Although Magic Sort has achieved strong revenue, Hexa Sort and other competitors are still growing, each game generates a stable revenue of 5 to 15 million US dollars per year.

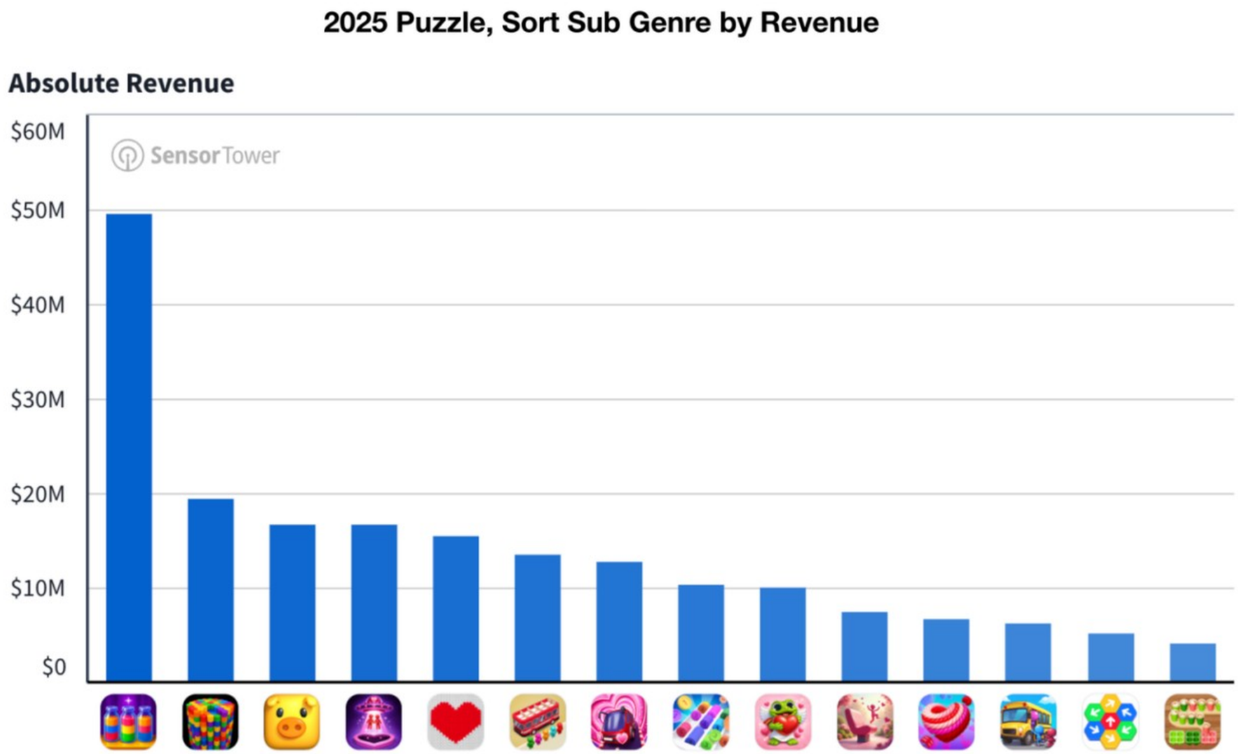
The segment has not become winner-takes-all. US players are willing to try several similar games, leaving room for differentiation through themes, containers, physical rules, and skins.

The segment has not become winner-takes-all.

Player Base

The players in the Sort category seem willing to play multiple games simultaneously, seeking novelty rather than being committed to a single product.

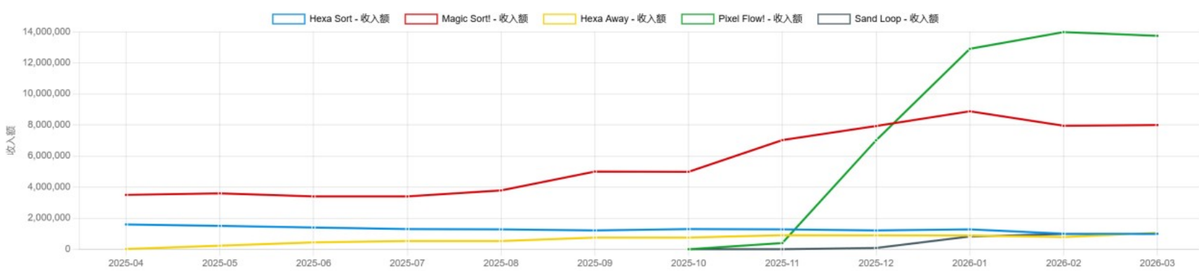
Such a market is unlikely to result in a winner-takes-all scenario and will definitely see a large number of copycat products or creative iterations.

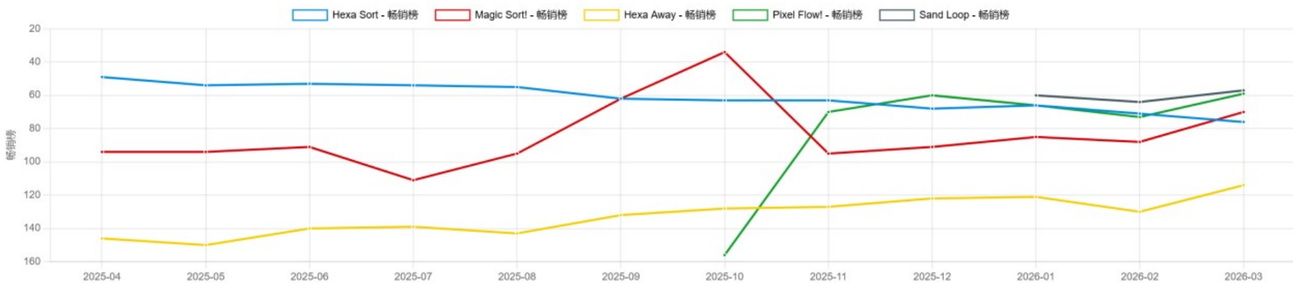


Sources: Naavik's《How Niche Subgenres are Shaping the Mobile Puzzle Market》

The Category Is Still Expanding

Five titles from the Water Sort category and variants, tracked 2025.04–2026.03 on downloads, revenue and grossing charts





Key Finding

A **"One-Strong, Many-Growing"** Expanding Market

- Flagship Magic Sort shows strong revenue power and growth momentum, proving the category's monetization ceiling
- Ranks 2–5 keep growing under the flagship's **siphon effect** — the market is expanding, not zero-sum
- Low loyalty + eagerness to try new titles (see US analysis) leaves ample room for entrants

Entering a market with proven demand at the top and a still-growing mid-tail
Differentiation (art / theme / side gameplay) captures incremental share

Sources: SensorTower and Appark (2025.04–2026.03 trends); category classification per SensorTower and Naavik

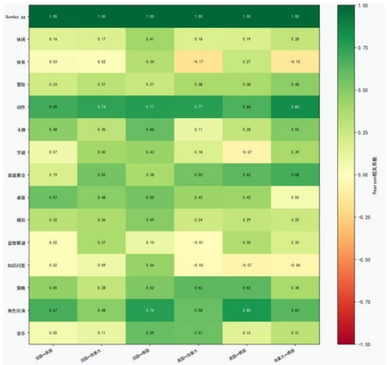


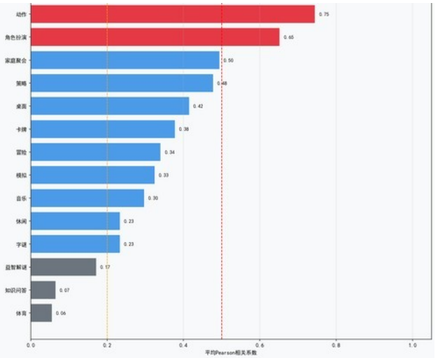
Module 4 | Cross-Market Trend Diffusion

Cross-Market Trend Transmission

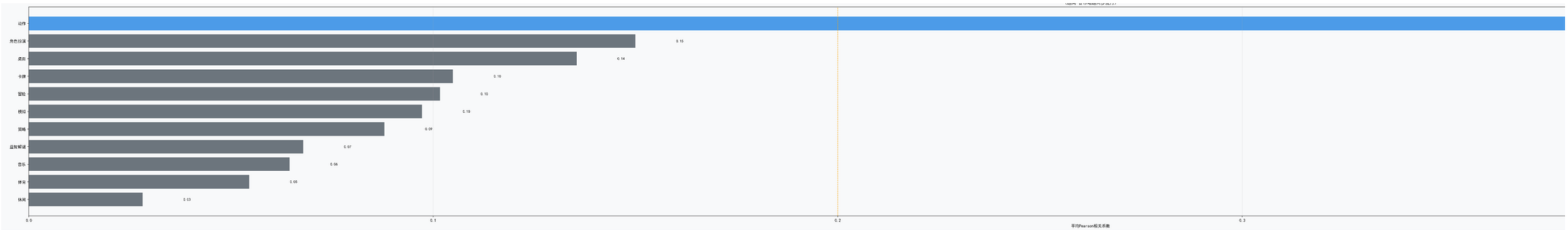
New products penetrate different social systems at different speeds, and **leading markets' consumption patterns tend to transmit to follower markets.** Aesthetics and demands within similar cultural circles are also fairly consistent.

Drawing 90-day popularity curves from each category's positions in each market's Top 40 free chart, and use the Pearson correlation coefficient to measure curve similarity between markets, we find that:





Canada's and Europe's game markets correlate most in category popularity, while Action games have the highest global universality.



Data source: Appark, etc.

Module 5 | User Profile Analysis

User Profile of Water Sort

Benchmarked against the category flagship product Magic Sort

Android

Female 18-24	Female 25-34	Female 35-44	Female 45-54	Female 55-99	Male 18-24	Male 25-34	Male 35-44	Male 45-54	Male 55-99
9.57%	19.04%	19.66%	10.06%	2.76%	8.41%	11.92%	11.76%	5.33%	1.49%

IOS

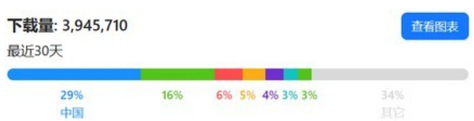
Female 18-24	Female 25-34	Female 35-44	Female 45-54	Female 55-99	Male 18-24	Male 25-34	Male 35-44	Male 45-54	Male 55-99
10.39%	18.86%	18.69%	9.77%	3.42%	8.20%	11.80%	11.35%	5.49%	2.03%

25~44 age
2:3 male-to-female ratio

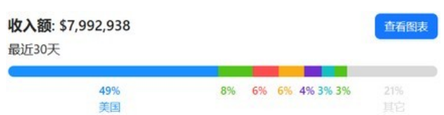
Economic independence and face considerable career and family pressure.
(In addition, the share of silver-haired players aged 55+ is extremely low)

D1 Retention	D2 Retention	D3 Retention	D4 Retention	D5 Retention	D6 Retention	D7 Retention	D14 Retention	D30 Retention
0.413163	0.329085	0.267609	0.224614	0.194759	0.173078	0.156626	0.100139	0.051826

Downloads & Revenue Sources



From left: China, the United States, Brazil, India, Germany, the United Kingdom, Egypt



From left: the United States, China, Germany, the United Kingdom, Canada, Australia, France

Open the game **3~5 times** per day
Each session lasting only **3.1~6.7 Minutes**

Played habitually during idle moments such as commuting, using the restroom, or waiting for the elevator.

User Esstential Needs

Based on Google Play reviews and word clouds (2026.03.29–04.11), **Ads are tolerable, but the Sense of Being Cheated is Not**

① Ad Problems

A 30s~3min ad every 1~2 levels; misclicks jump to the store and force a restart; ads persist even after paid ad-removal

→ The direct cause of most churn

② Paywall

Levels turn Hard/Super Hard at 60~100; failing disables pre-equipped boosters — re~purchase costs 900 coins (≈45 wins)

→ Widely read as "pay or fail"

③ Hidden-Color Luck

Late levels hinge on hidden colors in bottles — solving becomes guessing; colors reshuffle on retry, killing save-scum

→ The last straw for skill players

④ Technical Bugs

Ad crashes, lost progress, bottle levels not clearing, abnormal coin drain, no offline play

→ Accelerates the trust collapse



Pain concentrates on **Ads, Difficulty Paywalls, Luck Mechanics, and Stability**

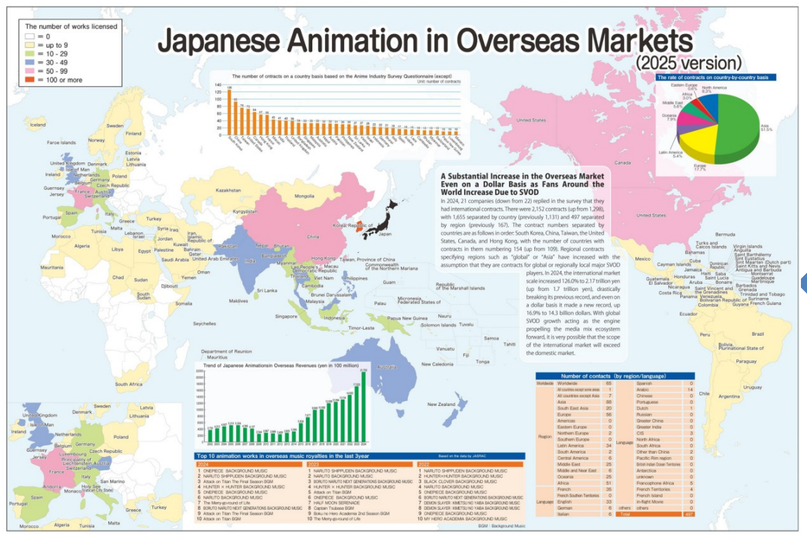
So we have to make a new title that is ad-restrained, skill-solvable, smoothly tuned and stable can absorb the churned, highly sticky users

Game Art Trends——Stylization

As rendering tech and AI generation advance rapidly, player aesthetics are shifting structurally

Positioning for Mini-Games

- Hyper-realism is unreachable in budget and schedule
- The way out = borrow from proven stylized works (Hades / Gris / Hollow Knight)
- Win niche segments with **differentiated art**





'Human Touches' Are Becoming the Key Attraction

Players are drawn to personality: characters' little gestures and quirks, tones reflecting unique life experience, creators' personal marks hidden in details and easter eggs.

The concurrent #RealLife #NoAI authenticity movement in the US further confirms this judgment.

Therefore, in order to differentiate ourselves from the style of our flagship products, we chose the second most popular style ——East Asian anime style for our sales.

Sources: Vsquar Art, Game Art Trends Shaping the Industry in 2026; BCG, Video Gaming Report 2026

17 Why Cats: The Highest Commercial Value



Under the subcategories of cartoon and animated comic styles, the elements of pets and animals, combined with collection systems such as Gacha and Gambling, have shown outstanding market performance.

① Macro Market

- 2025 global pet market **$346B**, +5.9–7.1%/yr; cats hold 30~35%

② User Fit

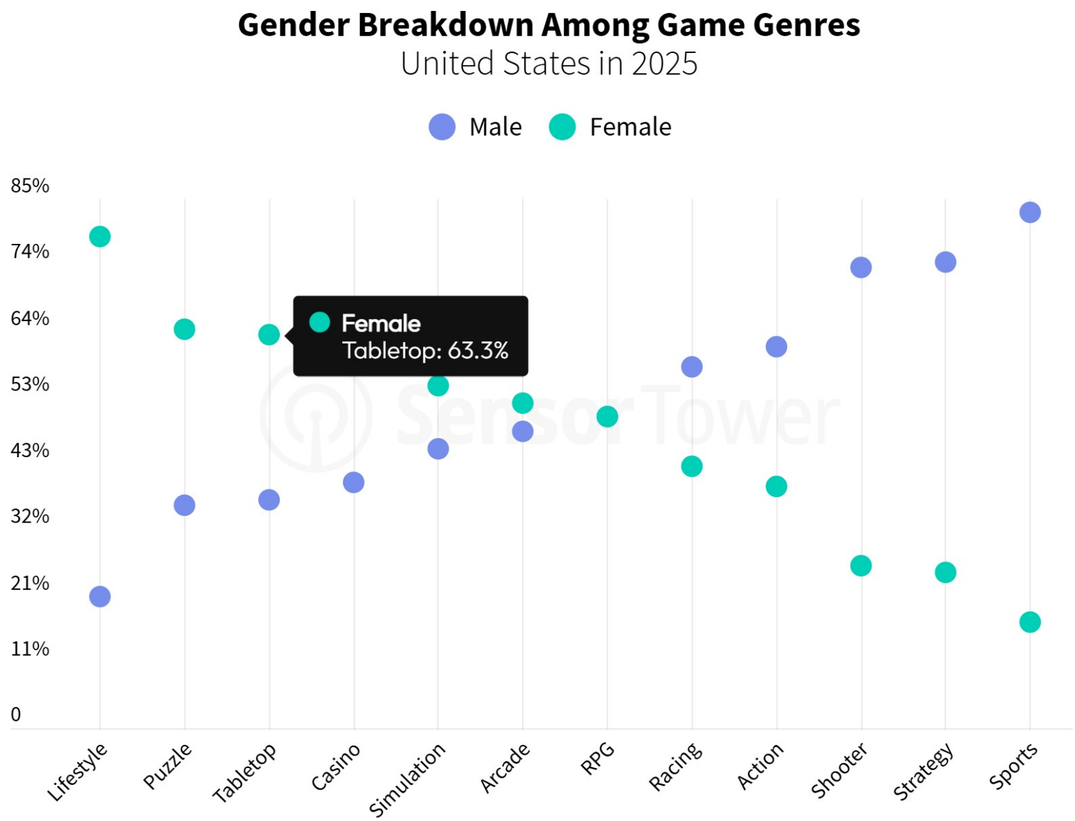
- Core cat spenders = **Gen Z and millennials**
- Matches the Sort core profile (25–44, female-leaning) precisely
- Search heat: dogs lead in the West, **cats lead in Asia**; cats win outright on YouTube

③ Market Proof

- Cat-element titles **outperform** within the same genres
- Cat IPs carry natural social spread and collection value

④ Style Gap

- Existing cat titles converge in art style — **cartoon/comic looks still open**
- Flat comic + animation style fits the big East-Asian markets
- A hand-polished feel lowers "AI-generated" negativity



Sources: Fortune Business Insights, Pet Care Market; Google Trends / Baidu Index; 2026 China Pet Industry White Paper; FindLaw; European Parliament

Side-Gameplay Candidates——The Big Three

Casual's three mainstream elements — Puzzle (match/merge), simulation, gambling mechanics (2025 full-year data)

品类标签	2024下载量	2025下载量	%	2024收入额	2025收入额	%	游戏排行榜
益智	7.0B	7.7B	9.9%	$7.6B	$8.2B	7.6%	
博彩	1.5B	1.7B	15.7%	$7.5B	$6.9B	-7.6%	
模拟	4.7B	4.9B	5.3%	$2.0B	$1.9B	-4.8%	
桌面游戏	1.9B	2.2B	14.1%	$870.4M	$933.0M	7.2%	
儿童	4.7B	5.1B	7.1%	$498.3M	$553.7M	11.1%	
冒险	767.9M	831.0M	8.2%	$574.5M	$542.3M	-5.6%	
街机	1.5B	1.7B	13.6%	$458.8M	$457.8M	-0.2%	
RPG	276.5M	251.2M	-9.2%	$522.6M	$424.1M	-18.8%	
派对游戏	849.5M	917.8M	8.0%	$325.0M	$272.5M	-16.2%	
体育类游戏	867.2M	845.8M	-2.5%	$282.1M	$257.5M	-8.7%	
策略	131.7M	207.8M	57.8%	$70.1M	$103.9M	48.2%	
动作	3.1B	3.3B	5.6%	$86.1M	$81.3M	-5.5%	
赛车	1.9B	1.9B	2.5%	$50.7M	$55.0M	8.6%	
射击	126.8M	94.3M	-25.7%	$3.9M	$2.7M	-30.7%	
游戏合集	214K	754K	253.1%	$21K	$94K	355.8%	
地理位置游戏	96K	156K	62.2%	$8K	$34K	346.7%	

Sources: Appmagic, 2026 Mobile App Market Outlook; SensorTower; Steam

Puzzle: Match-3 Still King, Merge Is Sprinting

- Puzzle earned **$8.2B+** in 2025 (+7.6%), downloads +9.9% — the only casual macro-genre still growing revenue
- Match-3 takes 58%+ of casual-puzzle revenue; Merge led by Gossip Harbor / Travel Town

Gambling: Strong Hook, Must Be 'Repackaged'

- In 2025, **50% of top casual titles** added light gambling mechanics in live-ops
- Direct gambling clashes with Sort and swings too wide > switch to **gacha**: the same Skinner-box variable-ratio reinforcement, but staked on in-game resources/characters, softening the gambling feel

Simulation: Stable Demand, Ossified Innovation

Farm leads at $955M (–2%); gameplay is fixed, profiting long-run on mature systems → fit as a **side-gameplay frame**, not the core

Screening logic 70% classic + 30% stable innovation, side gameplay must be accepted by core users and self-consistent with the Sort main loop

Restaurant Sim × Match-3 Gacha × Incentives

With Sort as the core, fold in concepts from simulation and match-3 to build a durable, long-term game loop

Sort Core Gameplay □ Restaurant Simulation □ Match-3 Gacha Collection

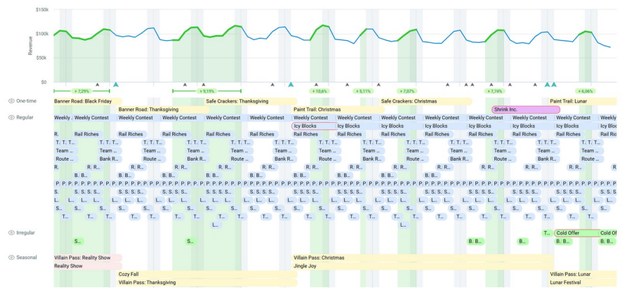
A Long-Term, Goal-Driven Core Loop

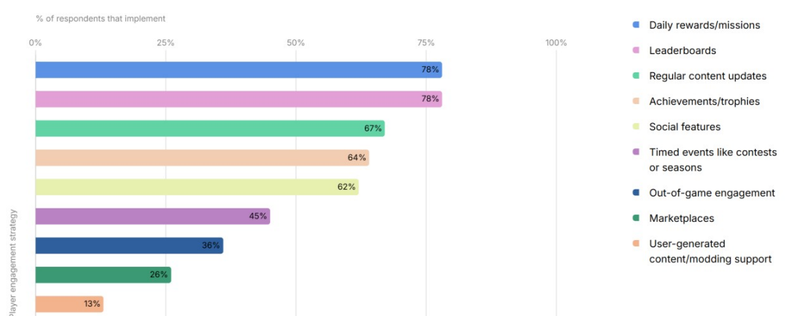
Restaurant Simulation

- An "order-driven" goal frame for Sort levels
- Shop growth gives mid/long-term nurturing goals
- Naturally coherent with sushi and cat themes

Match-3 × Gacha Collection

- Gacha = variable-ratio reinforcement, keeping the psychological hook
- Cat characters/skins as collection targets soften the gambling feel
- A pity system eases bad-luck frustration, fitting the "cat game" tone





[Source: Cint™ survey 2025, 300 respondents]

Incentives and Content

- Daily quests / check-ins / achievements sustain revisits
- Character progression carries resource output and sink
- Positive-dominant, controlled-negative feedback balance

Sorting Element —— Why Sushi?



Food themes are doubly universal in games and in life — game characters are both producers and consumers

- **Mechanic fit**

Sorting elements must be classifiable, identifiable, tactile — sushi toppings deliver.

- **Art fit**

Coherent with the cat theme and the clean, soft, fresh visual style.

- **Market fit**

Clearly differentiated from rivals' Western fast-food and liquid-sorting elements.

Widely used in nearly all kinds of games before.

YouGov
What the world thinks

YouGov - Global Cuisine Survey

	Country											
	Australia	China	Hong Kong	Indonesia	Japan	Malaysia	Philippines	Taiwan	Thailand	Vietnam	India	Singapore
Japanese cuisine												
Unweighted base	1020	1014	1023	1024	1000	1003	1019	1036	1020	1011	1163	1010
Base	1020	1014	1023	1024	1000	1003	1019	1036	1020	1011	1163	1010
Like a lot	33%	12%	57%	31%	67%	27%	47%	60%	48%	38%	13%	69%
Like a fair amount	32%	29%	35%	34%	24%	39%	38%	31%	38%	34%	24%	23%
Don't like very much	13%	22%	5%	14%	2%	16%	6%	7%	6%	10%	13%	4%
Don't like at all	5%	8%	1%	3%	0%	2%	1%	1%	2%	2%	6%	2%
Don't know	5%	5%	1%	8%	3%	6%	3%	1%	2%	4%	10%	1%
N/a - I have never eaten this cuisine	11%	24%	1%	10%	3%	11%	5%	1%	4%	11%	34%	2%
% of those who have ever eaten the cuisine that like it	74%	54%	93%	72%	94%	73%	90%	92%	90%	81%	57%	94%
American cuisine (i.e. from the USA)												
Unweighted base	1020	1014	1023	1024	1000	1003	1019	1036	1020	1011	1163	1010
Base	1020	1014	1023	1024	1000	1003	1019	1036	1020	1011	1163	1010
Like a lot	21%	7%	10%	21%	8%	12%	50%	26%	16%	21%	23%	27%
Like a fair amount	39%	24%	46%	34%	37%	37%	37%	44%	43%	32%	34%	46%
Don't like very much	16%	16%	23%	11%	14%	12%	3%	18%	11%	11%	9%	10%
Don't like at all	6%	6%	4%	3%	2%	2%	1%	2%	3%	2%	3%	2%
Don't know	7%	8%	7%	14%	18%	13%	3%	2%	8%	9%	8%	3%
N/a - I have never eaten this cuisine	12%	39%	10%	17%	20%	25%	6%	8%	18%	24%	24%	13%
% of those who have ever eaten the cuisine that like it	68%	51%	62%	66%	57%	65%	93%	76%	73%	71%	75%	83%
Mexican cuisine												
Unweighted base	1020	1014	1023	1024	1000	1003	1019	1036	1020	1011	1163	1010
Base	1020	1014	1023	1024	1000	1003	1019	1036	1020	1011	1163	1010
Like a lot	32%	5%	6%	6%	5%	8%	28%	11%	8%	10%	19%	17%
Like a fair amount	38%	15%	32%	19%	26%	26%	41%	32%	27%	20%	30%	38%
Don't like very much	12%	11%	19%	11%	7%	12%	6%	19%	13%	16%	9%	12%
Don't like at all	4%	6%	4%	4%	1%	3%	1%	2%	4%	4%	4%	3%
Don't know	4%	10%	8%	27%	21%	18%	6%	6%	13%	15%	8%	7%
N/a - I have never eaten this cuisine	10%	53%	31%	33%	40%	34%	19%	30%	34%	35%	29%	24%
% of those who have ever eaten the cuisine that like it	77%	42%	54%	38%	51%	51%	85%	62%	54%	46%	70%	72%



Sushi Cat Sort

Sort Core Gameplay × Cat Theme (East-Asian Anime Style) × Sushi Sorting Elements × Restaurant Sim × Gacha Collection